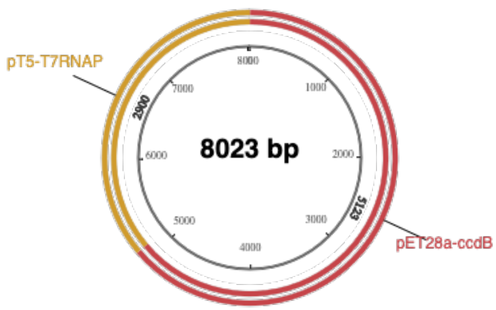
pET28a-pT5-T7RNAP

Created: 4/10/2025, 10:28:41 AM
Saved: 4/11/2025, 1:47:18 PM

Component Fragments

Name	Length	Produced by	5' End	3' End
pET28a-ccdB	5143	PCR	Fwd Primer (auto)	Rev Primer (auto)
pT5-T7RNAP	2920	PCR	Fwd Primer (auto)	Rev Primer (auto)



Notes

- Everything looks OK. No major issues detected.

Required oligos

Name	Primer 5' (overlap/spacer/ANNEAL) 3'	Len	%GC	3' %GC	3' Tm	3' Ta
pET28a-ccdB_fwd	tgtgcacctg**GCTTGATCCGGCTGCTAAC**	29	59	58	66.4	65.6
pET28a-ccdB_rev	tatgaggaca**ATTTCGCGGGATCGAGATC**	29	48	53	64.6	65.6
pT5-T7RNAP_fwd	cccgcgaaat**TGTCCTCATAGTTTGAACAAG**	31	45	38	59.2	60.2
pT5-T7RNAP_rev	cggatcaagc**CAGGTGCACATTAAGCTTG**	29	52	47	61.3	60.2

Build Settings

Property	Value
Product/Kit	#E5520 NEBuilder HiFi DNA Assembly Cloning Kit
Minimum Overlap	20 nt
Minimum Overlap Tm	48 °C
Circularize	Yes
PCR Polymerase/Kit	Q5 High-Fidelity DNA Polymerase
PCR Primer Conc.	500 nM
Min. Primer Length	18 nt

Assembled Sequence

```
#LOCUS       pET28a-pT5-T7RNAP    8023 bp  ds-DNA  circular   SYN  11-APR-2025
#DEFINITION  synthetic DNA
#ACCESSION   .
#VERSION     .
#KEYWORDS    NEBuilder
#SOURCE      synthetic DNA construct
#  ORGANISM  synthetic DNA construct
#REFERENCE   1  (bases 1 to 8023)
#  AUTHORS   .
#  TITLE     NEBuilder-generated Construct
#  JOURNAL   Exported  11-APR-2025 from NEBuilder https://nebuilder.neb.com
#COMMENT     NEBuilder-generated oligos (UPPERCASE = gene-specific, lowercase = overlap)
#COMMENT     pET28a-ccdB_fwd: tgtgcacctgGCTTGATCCGGCTGCTAAC
#COMMENT     pET28a-ccdB_fwd 3'Tm: 66.4  3'Ta: 65.6
#COMMENT     pET28a-ccdB_rev: tatgaggacaATTTCGCGGGATCGAGATC
#COMMENT     pET28a-ccdB_rev 3'Tm: 64.6  3'Ta: 65.6
#COMMENT     pT5-T7RNAP_fwd: cccgcgaaatTGTCCTCATAGTTTGAACAAG
#COMMENT     pT5-T7RNAP_fwd 3'Tm: 59.2  3'Ta: 60.2
#COMMENT     pT5-T7RNAP_rev: cggatcaagcCAGGTGCACATTAAGCTTG
#COMMENT     pT5-T7RNAP_rev 3'Tm: 61.3  3'Ta: 60.2
#FEATURES             Location/Qualifiers
#     source          1..8023
#                     /organism="synthetic DNA construct"
#                     /mol_type="other DNA"
#                     /plasmid="pET28a-pT5-T7RNAP"
#     gene          1..5123
#                     /note="pET28a-ccdB"
#     gene          5124..8023
#                     /note="pT5-T7RNAP"
#     primer_bind     8014..19
#                     /note="pET28a-ccdB_fwd"
#                     /note="gene-specific Tm: 66.4  Ta: 65.6"
#                     /note="overlap: tgtgcacctg"
#                     /note="gene-specific primer: GCTTGATCCGGCTGCTAAC"
#     primer_bind     complement(5105..5133)
#                     /note="pET28a-ccdB_rev"
#                     /note="gene-specific Tm: 64.6  Ta: 65.6"
#                     /note="overlap: tatgaggaca"
#                     /note="gene-specific primer: ATTTCGCGGGATCGAGATC"
#     primer_bind     5114..5144
#                     /note="pT5-T7RNAP_fwd"
#                     /note="gene-specific Tm: 59.2  Ta: 60.2"
#                     /note="overlap: cccgcgaaat"
#                     /note="gene-specific primer: TGTCCTCATAGTTTGAACAAG"
#     primer_bind     complement(8005..10)
#                     /note="pT5-T7RNAP_rev"
#                     /note="gene-specific Tm: 61.3  Ta: 60.2"
#                     /note="overlap: cggatcaagc"
#                     /note="gene-specific primer: CAGGTGCACATTAAGCTTG"
#ORIGIN
#        1 gcttgatccg gctgctaaca aagcccgaaa ggaagctgag ttggctgctg ccaccgctga
#       61 gcaataacta gcataacccc ttggggcctc taaacgggtc ttgaggggtt ttttgctgaa
#      121 aggaggaact atatccggat tggcgaatgg gacgcgccct gtagcggcgc attaagcgcg
#      181 gcgggtgtgg tggttacgcg cagcgtgacc gctacacttg ccagcgccct agcgcccgct
#      241 cctttcgctt tcttcccttc ctttctcgcc acgttcgccg gctttccccg tcaagctcta
#      301 aatcgggggc tccctttagg gttccgattt agtgctttac ggcacctcga ccccaaaaaa
#      361 cttgattagg gtgatggttc acgtagtggg ccatcgccct gatagacggt ttttcgccct
#      421 ttgacgttgg agtccacgtt ctttaatagt ggactcttgt tccaaactgg aacaacactc
#      481 aaccctatct cggtctattc ttttgattta taagggattt tgccgatttc ggcctattgg
#      541 ttaaaaaatg agctgattta acaaaaattt aacgcgaatt ttaacaaaat attaacgttt
#      601 acaatttcag gtggcacttt tcggggaaat gtgcgcggaa cccctatttg tttatttttc
#      661 taaatacatt caaatatgta tccgctcatg aattaattct tagaaaaact catcgagcat
#      721 caaatgaaac tgcaatttat tcatatcagg attatcaata ccatattttt gaaaaagccg
#      781 tttctgtaat gaaggagaaa actcaccgag gcagttccat aggatggcaa gatcctggta
```

```
#     841 tcggtctgcg attccgactc gtccaacatc aatacaacct attaatttcc cctcgtcaaa
#     901 aataaggtta tcaagtgaga aatcaccatg agtgacgact gaatccggtg agaatggcaa
#     961 aagtttatgc atttctttcc agacttgttc aacaggccag ccattacgct cgtcatcaaa
#    1021 atcactcgca tcaaccaaac cgttattcat tcgtgattgc gcctgagcga gacgaaatac
#    1081 gcgatcgctg ttaaaaggac aattacaaac aggaatcgaa tgcaaccggc gcaggaacac
#    1141 tgccagcgca tcaacaatat tttcacctga atcaggatat tcttctaata cctggaatgc
#    1201 tgttttcccg gggatcgcag tggtgagtaa ccatgcatca tcaggagtac ggataaaatg
#    1261 cttgatggtc ggaagaggca taaattccgt cagccagttt agtctgacca tctcatctgt
#    1321 aacatcattg gcaacgctac ctttgccatg tttcagaaac aactctggcg catcgggctt
#    1381 cccatacaat cgatagattg tcgcacctga ttgcccgaca ttatcgcgag cccatttata
#    1441 cccatataaa tcagcatcca tgttggaatt taatcgcggc ctagagcaag acgtttcccg
#    1501 ttgaatatgg ctcataacac cccttgtatt actgtttatg taagcagaca gttttattgt
#    1561 tcatgaccaa aatcccttaa cgtgagtttt cgttccactg agcgtcagac cccgtagaaa
#    1621 agatcaaagg atcttcttga gatccttttt ttctgcgcgt aatctgctgc ttgcaaacaa
#    1681 aaaaaccacc gctaccagcg gtggtttgtt tgccggatca agagctacca actctttttc
#    1741 cgaaggtaac tggcttcagc agagcgcaga taccaaatac tgtccttcta gtgtagccgt
#    1801 agttaggcca ccacttcaag aactctgtag caccgcctac atacctcgct ctgctaatcc
#    1861 tgttaccagt ggctgctgcc agtggcgata agtcgtgtct taccgggttg gactcaagac
#    1921 gatagttacc ggataaggcg cagcggtcgg gctgaacggg gggttcgtgc acacagccca
#    1981 gcttggagcg aacgacctac accgaactga gatacctaca gcgtgagcta tgagaaagcg
#    2041 ccacgcttcc cgaagggaga aaggcggaca ggtatccggt aagcggcagg gtcggaacag
#    2101 gagagcgcac gagggagctt ccagggggaa acgcctggta tctttatagt cctgtcgggt
#    2161 ttcgccacct ctgacttgag cgtcgatttt tgtgatgctc gtcagggggg cggagcctat
#    2221 ggaaaaacgc cagcaacgcg gcctttttac ggttcctggc cttttgctgg ccttttgctc
#    2281 acatgttctt tcctgcgtta tcccctgatt ctgtggataa ccgtattacc gcctttgagt
#    2341 gagctgatac cgctcgccgc agccgaacga ccgagcgcag cgagtcagtg agcgaggaag
#    2401 cggaagagcg cctgatgcgg tattttctcc ttacgcatct gtgcggtatt tcacaccgca
#    2461 tatatggtgc actctcagta caatctgctc tgatgccgca tagttaagcc agtatacact
#    2521 ccgctatcgc tacgtgactg ggtcatggct gcgccccgac acccgccaac acccgctgac
#    2581 gcgccctgac gggcttgtct gctcccggca tccgcttaca gacaagctgt gaccgtctcc
#    2641 gggagctgca tgtgtcagag gttttcaccg tcatcaccga aacgcgcgag gcagctgcgg
#    2701 taaagctcat cagcgtggtc gtgaagcgat tcacagatgt ctgcctgttc atccgcgtcc
#    2761 agctcgttga gtttctccag aagcgttaat gtctggcttc tgataaagcg ggccatgtta
#    2821 agggcggttt tttcctgttt ggtcactgat gcctccgtgt aagggggatt tctgttcatg
#    2881 ggggtaatga taccgatgaa acgagagagg atgctcacga tacgggttac tgatgatgaa
#    2941 catgcccggt tactggaacg ttgtgagggt aaacaactgg cggtatggat gcggcgggac
#    3001 cagagaaaaa tcactcaggg tcaatgccag cgcttcgtta atacagatgt aggtgttcca
#    3061 cagggtagcc agcagcatcc tgcgatgcag atccggaaca taatggtgca gggcgctgac
#    3121 ttccgcgttt ccagacttta cgaaacacgg aaaccgaaga ccattcatgt tgttgctcag
#    3181 gtcgcagacg ttttgcagca gcagtcgctt cacgttcgct cgcgtatcgg tgattcattc
#    3241 tgctaaccag taaggcaacc ccgccagcct agccgggtcc tcaacgacag gagcacgatc
#    3301 atgcgcaccc gtggggccgc catgccggcg ataatggcct gcttctcgcc gaaacgtttg
#    3361 gtggcgggac cagtgacgaa ggcttgagcg agggcgtgca agattccgaa taccgcaagc
#    3421 gacaggccga tcatcgtcgc gctccagcga aagcggtcct cgccgaaaat gacccagagc
#    3481 gctgccggca cctgtcctac gagttgcatg ataaagaaga cagtcataag tgcggcgacg
#    3541 atagtcatgc cccgcgccca ccggaaggag ctgactgggt tgaaggctct caagggcatc
#    3601 ggtcgagatc ccggtgccta atgagtgagc taacttacat taattgcgtt gcgctcactg
#    3661 cccgctttcc agtcgggaaa cctgtcgtgc cagctgcatt aatgaatcgg ccaacgcgcg
#    3721 gggagaggcg gtttgcgtat tgggcgccag ggtggttttt cttttcacca gtgagacggg
#    3781 caacagctga ttgcccttca ccgcctggcc ctgagagagt tgcagcaagc ggtccacgct
#    3841 ggtttgcccc agcaggcgaa aatcctgttt gatggtggtt aacggcggga tataacatga
#    3901 gctgtcttcg gtatcgtcgt atcccactac cgagatatcc gcaccaacgc gcagcccgga
#    3961 ctcggtaatg gcgcgcattg cgcccagcgc catctgatcg ttggcaacca gcatcgcagt
#    4021 gggaacgatg ccctcattca gcatttgcat ggtttgttga aaaccggaca tggcactcca
#    4081 gtcgccttcc cgttccgcta tcggctgaat ttgattgcga gtgagatatt tatgccagcc
#    4141 agccagacgc agacgcgccg agacagaact taatgggccc gctaacagcg cgatttgctg
#    4201 gtgacccaat gcgaccagat gctccacgcc cagtcgcgta ccgtcttcat gggagaaaat
#    4261 aatactgttg atgggtgtct ggtcagagac atcaagaaat aacgccggaa cattagtgca
#    4321 ggcagcttcc acagcaatgg catcctggtc atccagcgga tagttaatga tcagcccact
#    4381 gacgcgttgc gcgagaagat tgtgcaccgc cgctttacag gcttcgacgc cgcttcgttc
#    4441 taccatcgac accaccacgc tggcacccag ttgatcggcg cgagatttaa tcgccgcgac
#    4501 aatttgcgac ggcgcgtgca gggccagact ggaggtggca acgccaatca gcaacgactg
#    4561 tttgcccgcc agttgttgtg ccacgcggtt gggaatgtaa ttcagctccg ccatcgccgc
#    4621 ttccactttt tcccgcgttt tcgcagaaac gtggctggcc tggttcacca cgcgggaaac
#    4681 ggtctgataa gagacaccgg catactctgc gacatcgtat aacgttactg gtttcacatt
#    4741 caccaccctg aattgactct cttccgggcg ctatcatgcc ataccgcgaa aggttttgcg
#    4801 ccattcgatg gtgtccggga tctcgacgct ctcccttatg cgactcctgc attaggaagc
#    4861 agcccagtag taggttgagg ccgttgagca ccgccgccgc aaggaatggt gcatgcaagg
#    4921 agatggcgcc caacagtccc ccggccacgg ggcctgccac catacccacg ccgaaacaag
```

```
#     4981 cgctcatgag cccgaagtgg cgagcccgat cttccccatc ggtgatgtcg gcgatatagg
#     5041 cgccagcaac cgcacctgtg gcgccggtga tgccggccac gatgcgtccg gcgtagagga
#     5101 tcgagatctc gatcccgcga aattgtcctc atagtttgaa caagtccgat caggcttcgt
#     5161 ccgagcgttc gcaggtggag ctcataaaaa atttatttgc tttgtgagcg gataacaatt
#     5221 ataatagatt caattgtgag cggataacaa tttcacacag aattcattaa agaggagaaa
#     5281 ttaactatga gaggatcgca tcaccatcac catcacggat ccatgaacac gattaacatc
#     5341 gctaagaacg acttctctga catcgaactg gctgctatcc cgttcaacac tctggctgac
#     5401 cattacggtg agcgtttagc tcgcgaacag ttggcccttg agcatgagtc ttacgagatg
#     5461 ggtgaagcac gcttccgcaa gatgtttgag cgtcaactta aagctggtga ggttgcggat
#     5521 aacgctgccg ccaagcctct catcactacc ctactcccta agatgattgc acgcatcaac
#     5581 gactggtttg aggaagtgaa agctaagcgc ggcaagcgcc cgacagcctt ccagttcctg
#     5641 caagaaatca agccggaagc cgtagcgtac atcaccatta agaccactct ggcttgccta
#     5701 accagtgctg acaatacaac cgttcaggct gtagcaagcg caatcggtcg ggccattgag
#     5761 gacgaggctc gcttcggtcg tatccgtgac cttgaagcta agcacttcaa gaaaaacgtt
#     5821 gaggaacaac tcaacaagcg cgtagggcac gtctacaaga aagcatttat gcaagttgtc
#     5881 gaggctgaca tgctctctaa gggtctactc ggtggcgagg cgtggtcttc gtggcataag
#     5941 gaagactcta ttcatgtagg agtacgctgc atcgagatgc tcattgagtc aaccggaatg
#     6001 gttagcttac accgccaaaa tgctggcgta gtaggtcaag actctgagac tatcgaactc
#     6061 gcacctgaat acgctgaggc tatcgcaacc cgtgcaggtg cgctggctgg catctctccg
#     6121 atgttccaac cttgcgtagt tcctcctaag ccgtggactg gcattactgg tggtggctat
#     6181 tgggctaacg gtcgtcgtcc tctggcgctg gtgcgtactc acagtaagaa agcactgatg
#     6241 cgctacgaag acgtttacat gcctgaggtg tacaaagcga ttaacattgc gcaaaacacc
#     6301 gcatggaaaa tcaacaagaa agtcctagcg gtcgccaacg taatcaccaa gtggaagcat
#     6361 tgtccggtcg aggacatccc tgcgattgag cgtgaagaac tcccgatgaa accggaagac
#     6421 atcgacatga atcctgaggc tctcaccgcg tggaaacgtg ctgccgctgc tgtgtaccgc
#     6481 aaggacaagg ctcgcaagtc tcgccgtatc agccttgagt tcatgcttga gcaagccaat
#     6541 aagtttgcta accataaggc catctggttc ccttacaaca tggactggcg cggtcgtgtt
#     6601 tacgctgtgt caatgttcaa cccgcaaggt aacgatatga ccaaaggact gcttacgctg
#     6661 gcgaaaggta aaccaatcgg taaggaaggt tactactggc tgaaaatcca cggtgcaaac
#     6721 tgtgcgggtg tcgataaggt tccgttccct gagcgcatca agttcattga ggaaaaccac
#     6781 gagaacatca tggcttgcgc taagtctcca ctggagaaca cttggtgggc tgagcaagat
#     6841 tctccgttct gcttccttgc gttctgcttt gagtacgctg gggtacagca ccacggcctg
#     6901 agctataact gctcccttcc gctggcgttt gacgggtctt gctctggcat ccagcacttc
#     6961 tccgcgatgc tccgagatga ggtaggtggt cgcgcggtta acttgcttcc tagtgaaacc
#     7021 gttcaggaca tctacgggat tgttgctaag aaagtcaacg agattctaca agcagacgca
#     7081 atcaatggga ccgataacga agtagttacc gtgaccgatg agaacactgg tgaaatctct
#     7141 gagaaagtca agctgggcac taaggcactg gctggtcaat ggctggctta cggtgttact
#     7201 cgcagtgtga ctaagcgttc agtcatgacg ctggcttacg ggtccaaaga gttcggcttc
#     7261 cgtcaacaag tgctggaaga taccattcag ccagctattg attccggcaa gggtctgatg
#     7321 ttcactcagc cgaatcaggc tgctggatac atggctaagc tgatttggga atctgtgagc
#     7381 gtgacggtgg tagctgcggt tgaagcaatg aactggctta agtctgctgc taagctgctg
#     7441 gctgctgagg tcaaagataa gaagactgga gagattcttc gcaagcgttg cgctgtgcat
#     7501 tgggtaactc ctgatggttt ccctgtgtgg caggaataca agaagcctat tcagacgcgc
#     7561 ttgaacctga tgttcctcgg tcagttccgc ttacagccta ccattaacac caacaaagat
#     7621 agcgagattg atgcacacaa acaggagtct ggtatcgctc ctaactttgt acacagccaa
#     7681 gacggtagcc accttcgtaa gactgtagtg tgggcacacg agaagtacgg aatcgaatct
#     7741 tttgcactga ttcacgactc cttcggtacc attccggctg acgctgcgaa cctgttcaaa
#     7801 gcagtgcgcg aaactatggt tgacacatat gagtcttgtg atgtactggc tgatttctac
#     7861 gaccagttcg ctgaccagtt gcacgagtct caattggaca aaatgccagc acttccggct
#     7921 aaaggtaact tgaacctccg tgacatctta gagtcggact tcgcgttcgc gtaactgcag
#     7981 gcatcgcgtt cgcgtaactg cagccaagct taatgtgcac ctg
#//
#
```